

① Hough Circle Transform.

Given:

edge points

$$P_1 = (5, 4)$$

$$P_2 = (-1, 6)$$

Candidate center

$$C_1 = (2, 4)$$

$$C_2 = (4, 3)$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

for $C_1 = (2, 4)$

for P_1 :

$$d = \sqrt{(5-2)^2 + (4-4)^2}$$

$$d = \sqrt{9} = 3$$

for P_2 :

$$d = \sqrt{(-1-2)^2 + (6-4)^2}$$

$$= \sqrt{(-3)^2 + (2)^2}$$

$$= \sqrt{9+4} = \sqrt{13}$$

for $C_2 = (4, 3)$

for P_1 :

$$d = \sqrt{(5-4)^2 + (4-3)^2}$$

$$= \sqrt{1+1} = \sqrt{2}$$

for P_2 :

$$d = \sqrt{(-1-4)^2 + (6-3)^2} = \sqrt{25+9} = \sqrt{34}$$

②

edge point

$$P = (8, 6)$$

$$C = (4, 3)$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(3 - 4)^2 + (6 - 3)^2}$$

$$= \sqrt{16 + 9}$$

$$= \sqrt{25}$$

③

edge point

$$P = (8, 8)$$

$$C = (5, 4)$$

$$\text{radius, } r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(8 - 5)^2 + (8 - 4)^2}$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25}$$

